EXPERIMENT NO. 4

PORT CHARACTERISTICS OF MICROWAVE TEES

AIM: Study the working of the E-Plane Tee, H-Plane Tee & Hybrid Tee and verify its port Characteristics.

APPARATUS:

- Microwave source (Klystron)
- Isolator
- Frequency meter
- Variable attenuator
- Slotted line
- Tuneable Probe
- Detector mount
- Matched Terminations
- E-Plane Tee, H- Plane Tee, Hybrid Tee
- Wave guide stand
- Cables & accessories
- CRO and accessories.

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THEORY:

1. E-Plane Tee Junction:

An E-Plane Tee is a wave guide Tee in which the axis of its side arm is parallel to the E-field of the main arm (or) collinear arms shown in Figure 1. An E-Plane Tee is a voltage or series junction symmetrical about the central arm so that signal to be split up is fed from it or the signal to be combined are taken from it. If the collinear arms are symmetric about the side arm there are two different characteristics as shown.

1. The Propagation of E-field through the E-plane Tee when EM waves in TE10 mode enter the junction from side arm is shown in Figure 2. Bending of the electric field as it leaves the side arm causes a type of opposition to fields to be setup in the main arm. If the junction is completely symmetrical the waves of the opposite phase that leave through the collinear arms are equal in magnitude as shown in Figure 2.
2. The wave entering the junction through collinear arms is soon in Figure 3. Both waves are added at Port 3 since both are opposite phase at Port 1 and Port 2.
3. If the instantaneous field entering the collinear arms is equal in magnitude and phase, the net field is zero in side arm.

4. In general the resultant field leaving the Junction through side arm is proportional to the difference between the instantaneous fields that entered the Junction from opposite directions. Also the effective value of the field leaving through the side arm is proportional to the phase difference between the entering fields. Maximum energy leaves the side arm where the waves entering the Junction through two main are in phase opposition

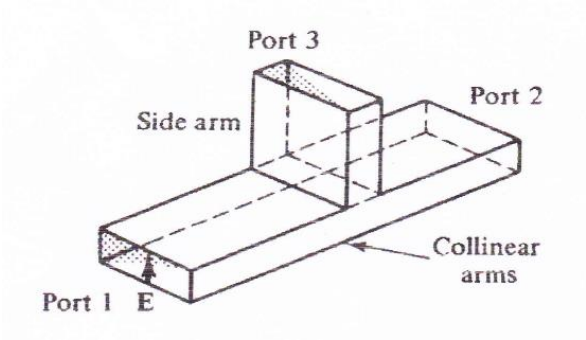

Figure 1. E-Plane Tee junction

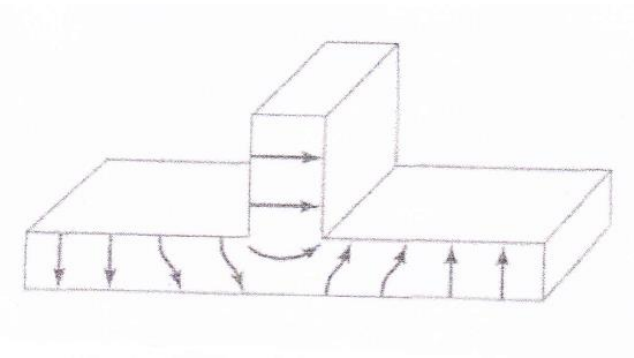

Figure 2. Distribution of the E-field

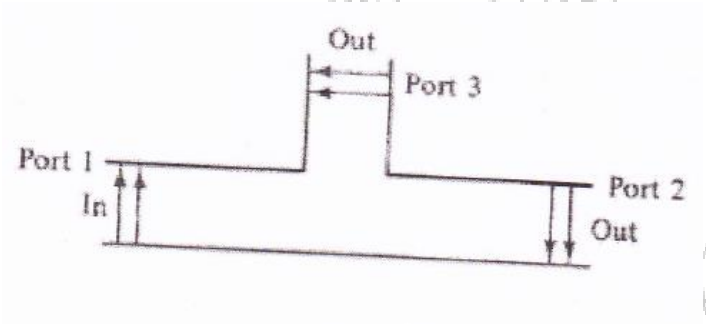

Figure 3. Field is added at Port 3

2. H-Plane Tee Junction:

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An H plane Tee is a wave guide Tee junction in which the axis of its side arm is shunting the E-field or parallel to the H-field of the main guide. As all three arms of H-plane Tee lie in the plane of magnetic field, the magnetic field divided itself into the arms. This is thus a current junction.

1. If the H-plane junction is completely symmetrical, and waves enter at Port 3 (side arm) leave through the collinear arms (port 1 and 2) with equal phase and magnitude. Since the electric field is not bent at the Junction but merely divides between two arms.
2. If two of same polarity are fed into port1 and port 2 of the collinear arm, the output wave of Port 3 will be in phase and additive.
3. If two input waves with equal magnitude and opposite phase are fed into Port 1 and Port 2, the output at Port 3 will be 0.

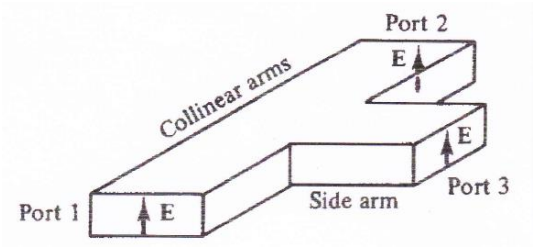


Figure 1. H-Plane Tee junction

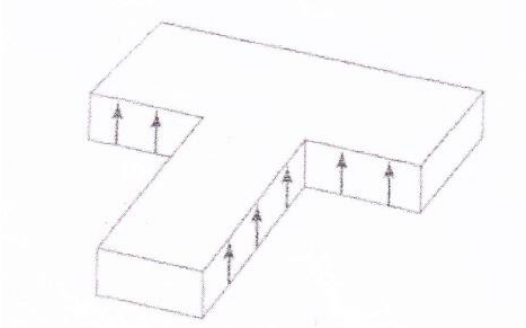


Figure 2. Distribution of the E-field

3. Magic or Hybrid Tee Junction:

A magic Tee is combination of an E-Plane and H-plane Tee. It is a 4 port hybrid circuit shown in Figure 1. A magic Tee is the most commonly form of hybrid circuit. The magic Tee is commonly used for mixing, duplexing and impedance measurements. The magic Tee has several characteristics, they are

1. If a wave is fed into Port 3 it will produce an output of equal magnitude and opposite phase at Port 1 and port 2 of the collinear arms and will not appear at Port 4. I.e. S43=0 (figure 2)
2. If a wave is fed into Port 4 (H arm)it will be divided equally between port 1 port 2 of the collinear arms and will not appear at Port 3 (E-arm) I.e S34=0 (figure 3)
3. If two waves of equal magnitude and same phase are fed into port1 and port 2, the output will be zero at Port 3 and additive at Port 4 (figure 4)
4. If a wave is fed into one of the collinear arm at Port 1 or port 2, it will not appear in the other collinear arm at Port 2 or port 1 because the E-arm causes a phase delay while the H-arm causes a phase advance I.e S12 =S21=0 (figure 5)
5. When waves of equal magnitude and phase enter the H-arm and E-arm the E-fields cancel one of the collinear arm and added in the other arm (figure 6)
6. If two waves of equal magnitude and opposite phase are fed into port1 and port 2 the output will be zero at Port 4 and additive at Port 3 (figure 7)

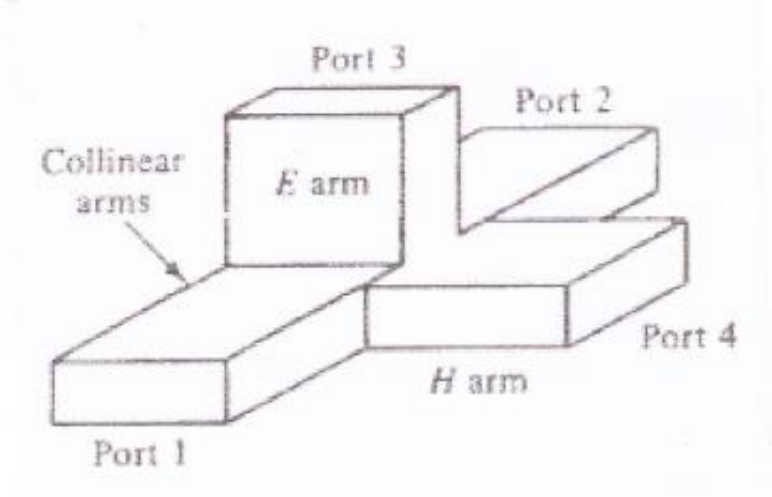
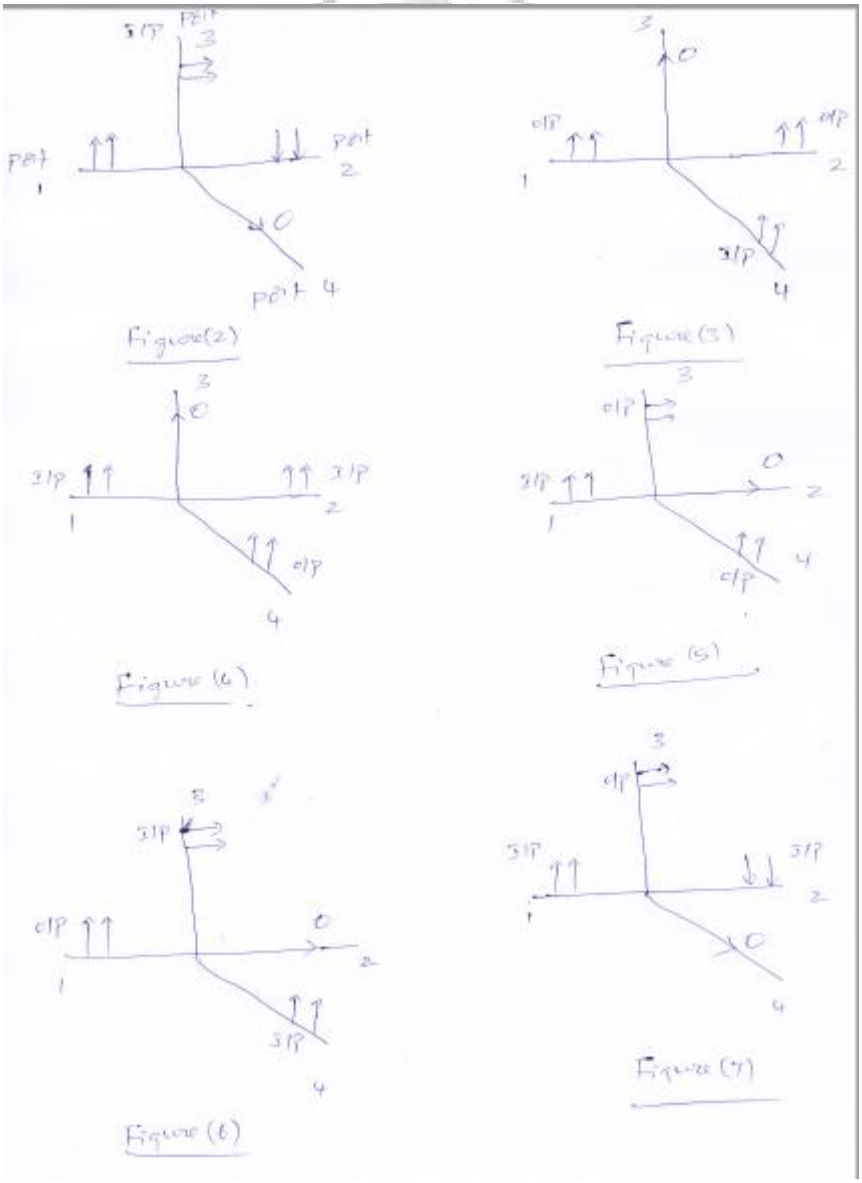

Port 3
Port 2
Collinear arms
E arm
Port 4
H arm
Port 1

Figure (1)


Figure(2)
Figure(3)
Figure (4)
Figure (5)
Figure (6)
Figure (7)

The basic parameters to be measured for magic Tee are defined below.

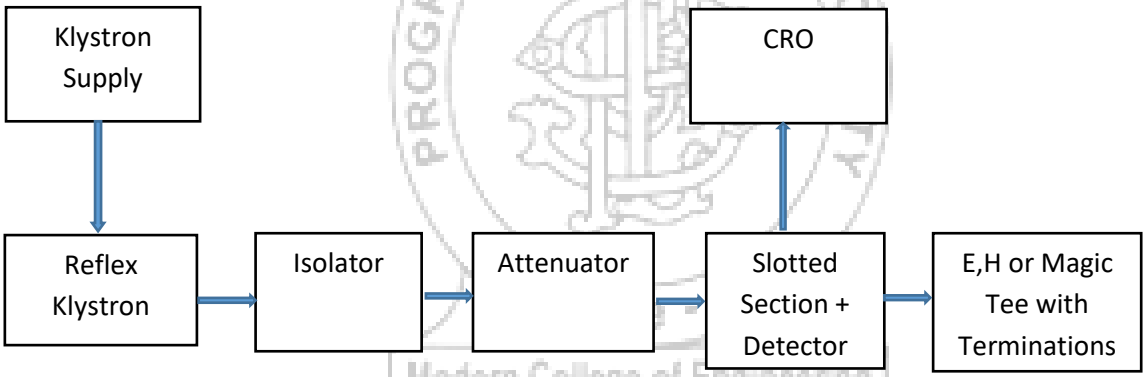
1. Input VSWR

   Value of SWR corresponding to each port, as a load to the line while other ports are terminated in matched load

2. Isolation

   The isolation between E and H arms is defined as the ratio of the power supplied by the generator connected to the E-arm (port 4) to the power detected at H -arm (port 3) when side arms I and 2 are terminated in matched load. The Isolation , in dB, is $10\log_{10}$ (P4/P3). Similarly, Isolation between other ports may also be defined.

PROCEDURE:



1. **VSWR Measurement of the Ports**

   a. Set up the components and equipment as shown in fig. keeping E arm towards slotted line and matched termination to other ports.

   b. Energize the microwave source for particular frequency of operation and tune the detector mount for maximum output.

   c. Measure the SWR of E-arm using the measured values of V_{max} and V_{min} with the help of slotted section and CRO.

   d. Connect another arm to slotted line and terminate the other port with matched termination. Measure the SWR as above. Similarly, SWR of any port can be measured.

2. Measurement of Isolation and Coupling Coefficient

a. Remove the tunable probe and Magic Tee from the slotted line and connect the detector mount to slotted line.

b. Energize the microwave source for particular frequency of operation and tune the detector mount for maximum output.

c. Observe the detected voltage shown on the CRO. Let it be P3.

d. Without disturbing the position of variable attenuator and gain control knob, carefully place the Magic Tee after slotted line keeping H-arm connected to slotted line, detector to E arm and matched termination to arm
1 and 2. Note down the reading of detected voltage on CRO. Let it be P4.

e. In the same way measure P1 & P2 by connecting detector on these ports one by one.

f. Determine the isolation between port 3 and 4 as P3-P4 in dB.

g. Determine the coupling coefficient by P3- P1 for port P1 & P2.

OBSERVATIONS: a) Beam Voltage = b) Beam current =

c) Repeller Voltage = d) Frequency = e) Output Voltage=

1. E-Plane Tee:

Input Port	Input Voltage	Output Voltage	
		Port1	Port2
E Port			

Table 1.

	Vmax (in *mV*)	Vmin (in *mV*)	VSWR
Port 1			
Port 2			
Port 3			

Table 2.

2. H-Plane Tee:

Input Port	Input Voltage	Output Voltage	
		Port1	Port2
H Port			

Table 3.

	Vmax (in *mV*)	Vmin (in *mV*)	VSWR
Port 1			
Port 2			
Port 3			

Table 4.

3. Magic Tee:

Input Port	Input Voltage	Output Voltage		
		Por1	Port2	E/H
H Port				
E Port				

Table 5.

	Vmax (in *mV*)	Vmin (in *mV*)	VSWR
Port 1			
Port 2			
Port 3			
Port 4			

Table 6.

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CALCULATIONS;

VSWR:

$$VSWR = \frac{V_{max}}{V_{min}}$$

RESULTS:

VSWR :

	VSWR for E Plane Tee	VSWR for H Plane Tee	VSWR for Magic Tee
Port 1			
Port 2			
Port 3			

CONCLUSION:

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QUESTIONS:

Q.1 What are the various type of Tees.
Q.2 What is H - plane Tee?
Q.3 What is E - plane Tee?
Q.4 What is Magic Tee?
Q.5 What is the electric property of H-plane Tee?
Q.6 What are the properties of E-plane Tee?

Q.7 What are the properties of Magic Tee?
Q.8 What are the applications of Magic Tee?
Q.9 What is the isolation between E & H arm?
Q.10 Define Coupling Coefficient?

ANSWERS:

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